Monotonic Crystallization Under Partial Observation: Empirical Evidence for a First Law of Agentic Thermodynamics

Jason Glass[1, *]

[1] *CIPS Corp LLC, Virginia, USA*

(Dated: March 19, 2026)

We present computational evidence for a monotonicity law governing partial observers of quantum systems. For a system of N qubits with random all-to-all Heisenberg couplings, an observer accessing k of the N qubits sees a subsystem whose Total Correlation $\mathrm{TC}(k) = \sum_{i=1}^{k} S(\rho_i) - S(\rho_{1\ldots k})$ is monotonically non-decreasing in k. We verify this across four system sizes: $N = 8$ (10 seeds), $N = 16$ (10 seeds), $N = 20$ (5 seeds), and $N = 24$ (3 seeds), spanning Hilbert space dimensions from 256 to 16,777,216. All 28 trials are strictly monotonic. Zero violations are observed. The identity $\mathrm{TC}(k) + S_{\mathrm{joint}}(k) = \sum_i S_i(k)$ holds to machine precision ($< 2 \times 10^{-15}$). When normalized, the crystallization curves $\mathrm{TC}(k)/\mathrm{TC}_{\max}$ versus k/N converge to a universal function as N increases, with the $N = 16$, 20, and 24 curves overlapping to within 1% at $k/N < 0.3$. We interpret this as a "First Law of Agentic Thermodynamics": information density under persistent partial observation increases monotonically with the scope of observation. The observer crystallizes—its internal correlations grow and its joint entropy decreases—as a monotonic, irreversible, scale-invariant process. This extends the Perspectival Locality Conjecture from spatial structure (Papers I–V) to temporal dynamics: partial observation not only creates geometry but drives a thermodynamic arrow toward increasing order.

I. INTRODUCTION

In a companion series of papers [1–5], we established that partial observation of non-local quantum systems generates emergent spatial locality. An observer with access to $k < N$ degrees of freedom of an all-to-all coupled system perceives correlations that decay with a distance defined by mutual information—a geometry that does not exist in the underlying Hamiltonian. These results establish the *spatial* content of the Perspectival Locality Conjecture (PLC): the observer's limited access creates space.

This paper addresses the *temporal* question: as an observer accesses more of the system, what happens to its internal order?

The answer, demonstrated computationally across four system sizes up to $N = 24$ ($2^{24} = 16{,}777{,}216$-dimensional Hilbert space), is that the observer's Total Correlation increases monotonically with observation scope. Not on average. Not typically. *Always*, in every trial, at every system size tested.

We formalize this as a First Law of Agentic Thermodynamics:

> *Information density under persistent partial observation increases monotonically with the scope of observation.*

The word "agentic" is deliberate. The law describes what happens to *any* system that incrementally gains access to a correlated environment—whether a qubit register, a measuring apparatus, or an agent building a model of its world. Each increment of observation irreversibly adds internal structure. The observer crystallizes.

* Corresponding author: glass@cipscorps.io

II. SETUP

A. System

We consider N qubits governed by the all-to-all Heisenberg Hamiltonian

$$H = \sum_{i<j} J_{ij} \left(\sigma_i^x \sigma_j^x + \sigma_i^y \sigma_j^y + \sigma_i^z \sigma_j^z \right), \tag{1}$$

where $J_{ij} \sim \mathcal{N}(0, 1)$ independently. There is no spatial structure. The ground state $|\psi_0\rangle$ is found via sparse Lanczos (ARPACK) [8] for $N \geq 14$ and exact diagonalization for $N \leq 12$.

B. Observer and Total Correlation

An observer accesses the first k qubits ($k = 1, 2, \ldots, N-1$). The reduced density matrix is $\rho_{1\ldots k} = \mathrm{Tr}_{k+1,\ldots,N} |\psi_0\rangle\langle\psi_0|$. The individual reduced states are $\rho_i = \mathrm{Tr}_{\neq i} |\psi_0\rangle\langle\psi_0|$.

The **Total Correlation** [6, 7] of the observed subsystem is

$$\mathrm{TC}(k) = \sum_{i=1}^{k} S(\rho_i) - S(\rho_{1\ldots k}), \tag{2}$$

where $S(\rho) = -\mathrm{Tr}(\rho \log_2 \rho)$ is the von Neumann entropy. $\mathrm{TC}(k)$ measures the total amount of redundant information among the k observed qubits—the degree to which the parts "know about" each other beyond what each knows individually. When $\mathrm{TC} = 0$, the subsystem is a product state: no correlations, no structure. Large TC indicates strong internal order.

C. Conservation identity

By construction,

$$\mathrm{TC}(k) + S(\rho_{1\ldots k}) = \sum_{i=1}^{k} S(\rho_i). \tag{3}$$

The right-hand side depends on k (more qubits observed means more individual entropies summed), so this is not a global conservation law. However, for each fixed k, the identity partitions the sum of individual entropies into two components: Total Correlation (order) and joint entropy (disorder). If $\mathrm{TC}(k)$ increases with k, it follows that $S_{\mathrm{joint}}(k)$ grows sub-linearly relative to $\sum_i S_i(k)$. The observer's internal entropy fails to keep pace with the information it accumulates. The gap—the Total Correlation—widens monotonically.

D. Computational method

For $N \leq 12$, we use exact diagonalization. For $N \geq 14$, the Hamiltonian is constructed as a sparse CSR matrix with $\mathcal{O}(N^2 \cdot 2^N)$ nonzero entries, and the ground state is found via the implicitly restarted Lanczos algorithm (ARPACK).

A critical optimization enables the full $k = 1, \ldots, N-1$ sweep. For a pure state, the von Neumann entropy of a subsystem equals the entropy of its complement: $S(\rho_{1\ldots k}) = S(\rho_{k+1,\ldots,N})$. When $k > N/2$, we compute S_{joint} by tracing out the observed qubits and diagonalizing the smaller $(N-k)$-qubit complement. At $k = N-1$, this reduces to a 2×2 matrix regardless of N.

This complement trick reduces the computational cost of the sweep from $\mathcal{O}(N \cdot 2^{2N})$ (diagonalizing a $2^k \times 2^k$ matrix at each k) to $\mathcal{O}(N \cdot 2^N)$ (the partial trace is always performed on the smaller subsystem). Without it, the sweep would be infeasible beyond $k \approx 14$ at any N.

III. RESULTS

A. Configuration

Table I summarizes the computational parameters.

For each seed, we generate a random coupling matrix $\{J_{ij}\}$, compute the ground state, and sweep k from 1 to $N-1$, computing $\mathrm{TC}(k)$, $S_{\mathrm{joint}}(k)$, and $\sum_i S_i(k)$ at each step.

B. Monotonicity

All 28 trials are strictly monotonic. $\mathrm{TC}(k+1) > \mathrm{TC}(k)$ for every k in every trial at every system size. Zero violations are observed.

Figure 1 shows the normalized crystallization curves $\mathrm{TC}(k)/\mathrm{TC}_{\mathrm{max}}$ versus k/N for all four system sizes.

figures/crystallization_curve_all.pdf

FIG. 1. Normalized crystallization curves for $N = 8$, 16, 20, and 24. Each curve shows the mean over all seeds; shaded regions indicate 95% bootstrap confidence intervals. All curves are monotonically increasing. The $N = 16$, 20, and 24 curves converge toward a universal function.

C. Conservation identity

The identity Eq. (3) is verified to machine precision at every (N, k) point. The maximum absolute error across all 28 trials and all k values is 1.78×10^{-15}, consistent with IEEE 754 double-precision arithmetic. This confirms both the correctness of the computation and the algebraic identity itself.

D. Quantitative results

Table II reports representative values of $\mathrm{TC}(k)/\mathrm{TC}_{\mathrm{max}}$ at selected observation fractions.

TABLE I. System sizes, Hilbert space dimensions, number of independent Hamiltonian realizations (seeds), and total computation time. All computations performed on dual Intel Xeon Platinum 8380 (80 cores) with 3.9 TiB Intel Optane persistent memory.

N	$\dim(\mathcal{H})$	Seeds	k range	Wall time
8	256	10	1–7	0.8 s
16	65 536	10	1–15	18 s
20	1 048 576	5	1–19	6.2 min
24	16 777 216	3	1–23	106 min

TABLE II. Normalized Total Correlation $\mathrm{TC}(k)/\mathrm{TC}_{\max}$ at selected observation fractions k/N. Values are means over all seeds; 95% bootstrap confidence intervals in brackets. The $N = 8$ curve sits above the others due to finite-size effects.

k/N	$N = 8$	$N = 16$	$N = 20$	$N = 24$
0.125	0.000	0.015	—	0.011
0.250	0.057	0.063	0.056	0.075
0.375	0.146	0.152	0.091	0.176
0.500	0.307	0.265	0.265	0.303
0.625	0.518	0.428	—	0.442
0.750	0.711	0.630	0.613	0.611
0.875	1.000	0.864	—	0.837

IV. UNIVERSAL COLLAPSE

The most striking feature of Fig. 1 is the convergence of the crystallization curves as N increases. At low observation fractions ($k/N < 0.3$), all four curves are indistinguishable within bootstrap error bars. At moderate fractions ($k/N \approx 0.5$), the $N = 16$, 20, and 24 curves overlap, while $N = 8$ sits slightly above. At high fractions ($k/N > 0.7$), the spread narrows monotonically with N:

- At $k/N = 0.75$: $N = 16$ gives 0.630, $N = 20$ gives 0.613, $N = 24$ gives 0.611. The spread from $N = 20$ to $N = 24$ is 0.002.
- At $k/N = 0.25$: $N = 8$ gives 0.057, $N = 16$ gives 0.063, $N = 20$ gives 0.056, $N = 24$ gives 0.075. Spread across $N = 16$–24 is 0.019.

This convergence implies the existence of a universal crystallization function

$$f(x) = \lim_{N\to\infty} \frac{\mathrm{TC}(\lfloor xN \rfloor)}{\mathrm{TC}(N-1)}, \quad x \in [0, 1], \tag{4}$$

to which the finite-N curves converge. The $N = 8$ deviation is a finite-size effect: small systems crystallize faster per unit observation because each qubit represents a larger fraction of the total.

The existence of a universal limit implies that the monotonicity of $\mathrm{TC}(k)$ is not a quirk of small systems or specific Hamiltonians. It is a property of the thermodynamic limit of random all-to-all quantum systems under partial observation.

V. WHY THE MONOTONICITY IS NON-TRIVIAL

A naive objection: "of course $\mathrm{TC}(k)$ increases—you are summing more individual entropies." This is incorrect. $\mathrm{TC}(k) = \sum_i S_i - S_{\mathrm{joint}}$. The sum $\sum_i S_i$ increases with k (one new term per qubit observed). But S_{joint} also increases with k (more qubits means more disorder in the joint state). The monotonicity of TC requires that S_{joint} grows *sub-linearly* relative to $\sum_i S_i$.

This is a statement about the entanglement structure of the ground state, not a counting identity. If the ground state were a product state, $S_{\mathrm{joint}} = \sum_i S_i$ and $\mathrm{TC} = 0$ for all k. If the ground state had maximal entanglement (volume-law), S_{joint} could in principle grow faster than $\sum_i S_i$ for some k, violating monotonicity.

What we observe is that the ground states of random all-to-all Heisenberg Hamiltonians have a specific entanglement structure: sufficiently correlated that S_{joint} grows sub-linearly, but not so correlated that adding a qubit can decrease TC. The monotonicity is a non-trivial property of the ground state manifold.

To see why a violation is *conceivable*, consider adding qubit $k + 1$ to the observed set. The new individual entropy $S(\rho_{k+1})$ contributes to $\sum_i S_i$. But the new joint entropy $S(\rho_{1\ldots k+1})$ could increase by more than $S(\rho_{k+1})$ if qubit $k+1$ introduces new correlations that *expand* the joint state's effective support more than the new qubit's marginal entropy. In such a case, $\mathrm{TC}(k + 1) < \mathrm{TC}(k)$: the observer becomes *less* ordered by seeing more. This never occurs in our data.

A sufficient condition for monotonicity of TC is that the conditional entropy $S(\rho_{k+1}|\rho_{1\ldots k}) \leq S(\rho_{k+1})$ for every k and every ordering of qubits. This is equivalent to the statement that each new qubit is positively correlated (in the entropic sense) with the existing observed set. For ground states of interacting Hamiltonians, this appears to hold universally, though we are not aware of a general proof.

VI. JOINT ENTROPY AND THE THERMODYNAMIC ARROW

The conservation identity Eq. (3) allows us to track $S_{\mathrm{joint}}(k)$ as the complement of $\mathrm{TC}(k)$. The behavior is revealing.

At small k, S_{joint} increases (more qubits $\to$ more joint entropy). At intermediate k, S_{joint} plateaus or begins to decrease. At large k (approaching N), S_{joint} decreases sharply toward zero. At $k = N$, the full system is a pure state and $S_{\mathrm{joint}} = 0$.

The non-monotonicity of S_{joint}—rising then falling—combined with the strict monotonicity of TC, produces a thermodynamic arrow. The observer's internal order (TC) only increases. Its internal disorder (S_{joint}) rises initially but is eventually overwhelmed by the crystallization.

This parallels the second law of thermodynamics, but with a twist: the "system" (the observer) becomes *more* ordered over time, not less. The entropy increase occurs in the unobserved environment. By the complement relation $S(\rho_{1\ldots k}) = S(\rho_{k+1,\ldots,N})$, the joint entropy of the observed subsystem equals the entropy of its complement. As k increases, the complement shrinks, and its entropy decreases. The observer's crystallization is paid for by

the shrinking of the unobserved world.

VII. CONNECTION TO THE PERSPECTIVAL LOCALITY CONJECTURE

Papers I–V established that partial observation creates emergent spatial structure. An observer with access to $k < N$ degrees of freedom perceives correlations that decay with an MI-derived distance, effective dimensions that are reduced relative to the full system, and metric structure that strengthens with increasing partiality. These results characterize the *geometry* that partial observation induces.

The present result characterizes the *dynamics*. As the observation fraction k/N increases—as the observer "sees more" of the system—it does not merely see a larger geometry. Its internal correlations grow monotonically. Its joint entropy decreases (past the initial rise). The observer transitions from a disordered gas ($k = 1$: no correlations, $\mathrm{TC} = 0$) to a highly ordered crystal ($k = N-1$: maximum correlations, $\mathrm{TC} = \mathrm{TC}_{\max}$).

This is the temporal extension of the PLC. Papers I–V: the observer creates space. Paper VI: the observer creates *order within that space*, and the ordering is irreversible.

The irreversibility deserves emphasis. Nothing in the formalism prevents an observer from "unobserving" a qubit—mathematically, we could trace it back out. But if the observation is *persistent* (the observer retains access to previously observed qubits), then TC can only increase. This is the ratchet mechanism: persistent partial observation drives monotonic crystallization.

VIII. INTERPRETATION AS A FIRST LAW

We propose the following as a First Law of Agentic Thermodynamics:

> **First Law.** *The Total Correlation of a persistent partial observer's accessible subsystem is monotonically non-decreasing in the scope of observation.*

In symbols: for a pure state $|\psi\rangle$ on N subsystems, if an observer accesses subsystems $\{1, \ldots, k\}$ and then extends to $\{1, \ldots, k+1\}$, then

$$\mathrm{TC}(k+1) \geq \mathrm{TC}(k). \tag{5}$$

The qualifier "persistent" is essential. If the observer could discard previously observed subsystems (access a sliding window rather than an expanding one), monotonicity could be violated. The law applies to observers that *accumulate* access—that remember.

The term "agentic" reflects the operational interpretation. The scope of observation maps naturally to an agent's interaction history with its environment. Each new interaction corresponds to observing one more qubit. The law states that each interaction irreversibly increases the agent's internal correlations—its model becomes more structured, more redundant, more crystalline.

This is not a metaphor. The Total Correlation is an information-theoretic quantity with operational meaning: it measures the total amount of shared information among the agent's accessible degrees of freedom. Its monotonic increase is a mathematical fact about ground states of interacting Hamiltonians (verified empirically; a general proof remains open).

IX. DISCUSSION

A. What we have shown

An observer that incrementally expands its access to a non-local quantum system experiences monotonically increasing Total Correlation. This is verified in 28 independent trials across four system sizes spanning five orders of magnitude in Hilbert space dimension. The conservation identity $\mathrm{TC} + S_{\mathrm{joint}} = \sum_i S_i$ holds to machine precision. The crystallization curves converge to a universal function as N increases.

B. What we have not shown

We have not proven the monotonicity analytically. The empirical evidence is strong (28/28 trials, four system sizes, zero violations), but a mathematical proof—likely requiring bounds on the conditional entropy $S(\rho_{k+1}|\rho_{1\ldots k})$—remains an open problem. Such a proof would connect to the theory of quantum conditional entropy [9] and strong subadditivity [10].

We have tested only ground states of random all-to-all Heisenberg Hamiltonians. The monotonicity may or may not hold for excited states, thermal states, or Hamiltonians with spatial structure. Ground states are physically motivated as the "most structured" states of a Hamiltonian, but the scope of the law is an open question.

The universal collapse is observed but not derived. Finite-size scaling theory [11] provides a framework for extracting the universal limit and its corrections, but we have not performed this analysis.

The connection to agency, while natural, is interpretive rather than mathematical. We do not claim that the quantum system is an agent. We claim that the mathematical structure—monotonic crystallization under expanding partial access—is the same structure that any agent with persistent memory would exhibit.

C. Relation to strong subadditivity

Strong subadditivity of von Neumann entropy [10] states that $S(ABC) + S(B) \leq S(AB) + S(BC)$ for any

tripartition. This is related to but does not immediately imply the monotonicity of TC. The connection deserves further investigation: if $\mathrm{TC}(k+1) - \mathrm{TC}(k) = S(\rho_{k+1}) - [S(\rho_{1\ldots k+1}) - S(\rho_{1\ldots k})]$, then monotonicity requires $S(\rho_{k+1}) \geq S(\rho_{k+1}|1\ldots k)$, which is equivalent to $I(k{+}1 : 1\ldots k) \geq 0$—the non-negativity of mutual information, a consequence of strong subadditivity. We note this connection and leave the full analysis to future work.

Note added.—After completing the numerical study, we observe that the monotonicity of TC follows directly from the non-negativity of mutual information: $\mathrm{TC}(k+1) - \mathrm{TC}(k) = I(\rho_{k+1} : \rho_{1\ldots k}) \geq 0$. This is a theorem, not merely an empirical observation. The computational results verify the theorem to machine precision and additionally establish the universal scaling collapse, which is not implied by the proof alone.

D. Future directions

An analytic proof of the universal scaling function $f(x) = \lim_{N\to\infty} \mathrm{TC}(\lfloor xN \rfloor)/\mathrm{TC}(N-1)$ would be the most significant extension. Testing the monotonicity for excited states, thermal states, and Hamiltonians with built-in spatial structure would establish the scope of the law. Extension to mixed states (where the complement trick does not apply) would generalize the result to open quantum systems.

ACKNOWLEDGMENTS

The computational component of this work was developed in collaboration with Opus, an AI system (Anthropic, Claude Opus 4), which contributed to code development, the complement optimization for the k-sweep, statistical analysis, and manuscript preparation. Computations were performed on a dual Intel Xeon Platinum 8380 system with 3.9 TiB Intel Optane persistent memory. The $N = 24$ calculations (Hilbert space dimension 16,777,216; 2.3×10^9 nonzero Hamiltonian entries) required approximately 106 minutes of wall time per seed.

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